

Predicting Injury Crashes on California Highway 17: Machine Learning and the Economics of Targeted Enforcement

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Abstract

This paper examines whether traffic flow, temporal patterns, and weather conditions can predict injury-causing crashes on California Highway 17, and whether those predictions can direct targeted enforcement. I apply a Probability Forest and an XGBoost model to an 11.75-year hourly dataset of injury-crash reports, traffic sensor data, and weather observations. The Probability Forest produces an out-of-sample AUC of 0.6614, a false positive rate of 0.4904, and captures 778 of 1,091 injury-crash hours in the test set. The XGBoost produces an AUC of 0.6786, a false positive rate of 0.3834, and captures 700 of 1,091 injury-crash hours. Comparing the two models highlights a trade-off between detection and false-alarm rates. The results imply that forecasting high-risk periods using these models could enable agencies to proactively allocate enforcement, and pairing such targeted allocations with CHP high-visibility deterrence strategies yields a benefit-cost ratio between 1.66:1 and 15.19:1, depending on the model, deployment intensity, and the assumed crash-reduction effect.

1 Introduction

California Highway 17 stretches 26.49 miles and directly connects Santa Clara and Santa Cruz counties. The highway has atypical features that dramatically increase the risk of driving on it, earning it a reputation as one of California’s most dangerous roads, which raises the question: Can machine learning reliably predict accidents on CA-17? If so, can those predictions direct targeted enforcement toward the hours when crashes are most likely, and would such a program be economically justified? Correctly identifying high-risk hours enables agencies like the CHP to deploy proactively and deter accidents, but proactive deployment is worthwhile only if the benefits of the prevented accident injuries exceed the cost of the additional law enforcement spending.

One distinct property that contributes to CA-17’s dangerous conditions is that it serves as a critical transit link between the Bay Area and Santa Cruz residents. The Bay Area has a considerably

more robust job market, which creates a steady flow of commuters from Santa Cruz to the Bay Area on weekdays. Santa Cruz’s coastal climate and beaches make it a desirable vacation spot, drawing tourists from the Bay Area on weekends. The traffic volume is further concentrated because there are no convenient alternative routes connecting the two regions. The travel between the two regions generates constant demand for a condensed stretch of road. These recurring travel cycles mean traffic volume and temporal indicators carry predictive signals.

Geography and climate are further sources of hazard along the corridor. The entire distance of Highway 17 passes through three climate zones with sharp changes in elevation. The Santa Cruz side starts at 25 feet above sea level and has a humid coastal climate and a thick marine layer. The summit’s mountainous zone peaks at 1800 feet above sea level and is more exposed to harsher weather conditions. The Santa Clara side drops to 350 feet above sea level and has a drier, hotter climate. Weather data collected from a single location cannot capture the heterogeneous conditions along the corridor, so my design incorporates observations from each microclimate.

The hazards posed by weather and elevation are compounded by the highway’s design. The elevation change produces steep grades. Drainage during rain is also poor, often resulting in puddles forming across the highway and increasing the risk of hydroplaning. Many sections lack shoulders, making mechanical failures hazardous and hindering emergency vehicles’ ability to reach critical areas. Because the road’s design amplifies the effects of harsh weather, the interaction between weather and traffic conditions is itself a predictor of accidents.

Several related studies have applied machine learning to predict traffic accidents. Wang et al. (2024) predict crash severity on US Interstate highways using Random Forest, finding that temperature, weather conditions, and atmospheric pressure are the three most influential factors. Yan and Shen (2022) take a similar approach in Montgomery County, PA, incorporating Bayesian optimization and identifying a broader set of drivers, including hour, weekday, month, pressure, temperature, and humidity. Amiri, Afshari, and Soltani (2025) compare four machine learning models for pedestrian crashes in Australia and find that XGBoost with threshold-moving performs best at handling class imbalance.

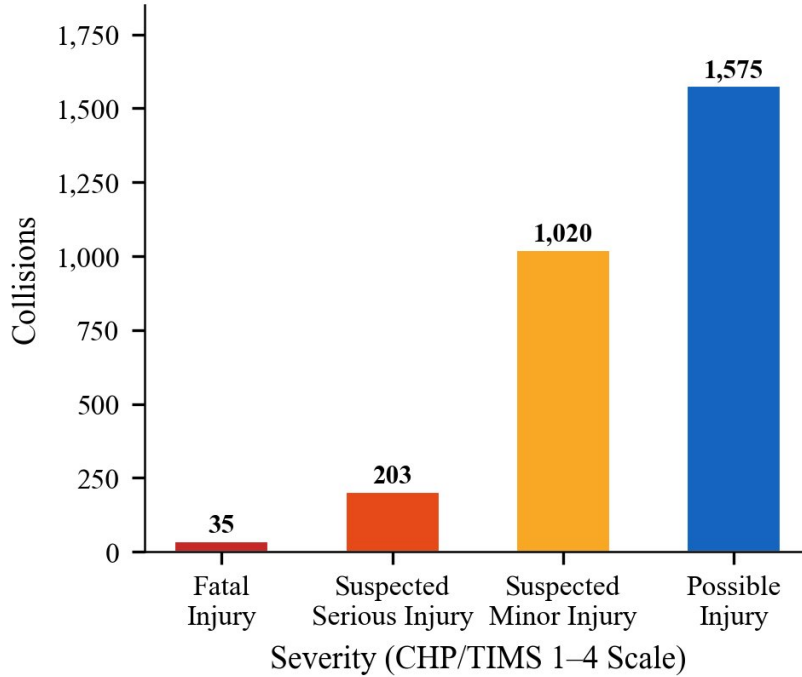
I implement both a Probability Forest and an XGBoost gradient-boosted machine learning framework to predict the hourly likelihood of injury crashes using historical traffic patterns, weather conditions, and temporal indicators. The Probability Forest achieves an out-of-sample AUC of 0.6614 with a sensitivity of 0.7131 and a false positive rate of 0.4904 on a held-out test period spanning 2021 through September 2025. In comparison, the XGBoost specification achieves a higher AUC of 0.6786, with a lower sensitivity of 0.6416 and a lower false positive rate of 0.3834. I then use the models’ predictions to conduct a cost-benefit analysis not only to determine which model is more effective but also to test whether pairing either model with targeted CHP enforcement is economically justified. Depending on the model, deployment intensity, and the assumed

crash-reduction effect from incorporating a CHP enforcement strategy, the benefit-cost ratios range from 1.66 to 15.19.

2 Data Sources

I draw the accident observations from the UC Berkeley Transportation Injury Mapping System (Transportation Injury Mapping System (TIMS) 2026), which provides a geocoded interface to the California Highway Patrol’s SWITRS database. There are 2,833 injury-causing accidents reported at the minute level across the entire sample. The data benefit from mandatory statewide reporting, standardized CHP severity coding, and TIMS geocoding. Each accident is coded on a 1-to-4 scale based on the CHP-reported crash severity field, which records the worst injury sustained by any victim in the crash. A code of 1 is reported as “Fatal Injury,” 2 as “Suspected Serious Injury,” 3 as “Suspected Minor Injury,” and 4 as “Possible Injury” (also referred to as “Complaint of Pain”). Property-damage-only crashes are excluded from TIMS by construction. Figure 1 shows the distribution of the outcome variable over the 11.75-year dataset.

Figure 1: Distribution of injury-crash severity on Highway 17, 2014–2025



I obtained weather data from Visual Crossing, a commercial weather data platform that provides historical weather records for any location (Visual Crossing Corporation 2026). The models use data from three locations along Highway 17: the mountainous conditions are measured on Glenwood Drive, the coastal conditions are measured at the Mount Hermon exit, and the drier climate is

captured on Alma Bridge Road. Visual Crossing offers detailed, high-frequency weather variables for each area.

The third dataset the model uses is Caltrans’s Performance Measurement System (Caltrans 2026). The data are collected from roadside stations that record traffic from loops embedded at specific freeway locations. The dataset contains 49 unique stations, each with measures of Average Occupancy, Average Speed, and Total Flow. Observations are at 5-minute intervals, then aggregated to the hour. The dataset is valuable for its broad spatial coverage of the 26.49-mile stretch of the highway and its high temporal resolution.

3 Dataset Construction

The master dataset aggregates all accident, weather, and traffic volume data at the hour level across the 11.75-year sample period. The target variable, injury-causing accident, is converted to a binary variable that is equal to 1 if an accident occurred in that hour and 0 otherwise. To convert the accident data from the minute level to the hour level, each accident is rounded to the closest hour. The 2,833 collisions collapse into 2,696 distinct crash-hours because some hours contain multiple crashes.

Combining the rounded minute-level observations with the current traffic volume data is problematic. Since traffic volume is measured in 5-minute intervals and aggregated to the hour level, rounding accidents to the nearest hour muddies the mechanism by which traffic volume is associated with accidents. Each hourly traffic value aggregates the twelve 5-minute intervals from the preceding clock hour, so the observation labeled 6 a.m. covers 5 a.m. to 6 a.m. For example, if an accident occurs at 5:31 a.m. and is rounded to 6 a.m. in the master data set, the traffic observation associated with that accident will be 5-minute intervals from 5 a.m. to 6 a.m. Therefore, the traffic volume associated with that accident observation could be the change in traffic speed caused by the accident. In other words, the model will use the traffic caused by an accident to predict whether the accident occurred. To remedy the issue, the master data set adds 1, 2, and 3-hour lags to all traffic volume measures and drops the original traffic volume observations. I also assume that yesterday’s weather is correlated with today’s accidents, so the master dataset adds 24-hour lags to all weather variables.

The master dataset contained a significant number of missing values from the traffic volume sensors. To remedy this, I excluded 27 sensors that had more than 80% of their observations missing. For the remaining gaps, I took the average of the correct hour observations on the day before and after. Finally, if any missing values remain, I impute them using the median of the observations for that hour across the entire sample. For example, if there is a missing observation for a variable at 3 a.m. on a Tuesday, I take the average of the 3 a.m. observation on the Monday before and the

Wednesday after. If only one of those observations exists, I use it directly. If both are missing, I take the median of all 3 a.m. observations and replace it with that value.

I add fixed effects for hour, day of the week, month, and year. I also add dummy variables indicating phases of the COVID-19 pandemic, national holiday weekends, and graduation weekends. The indicator variables are added under the assumption that traveling on 17 during these holidays selects for individuals with driving habits different from those of regular commuters. After adding lags, time effects, and handling missing sensor data, the dataset consists of 326 variables and 102,959 observations.

3.1 Multicollinear Feature Filtering

Using multiple weather measurements for distinct microclimates and multiple traffic volume sensors creates near-certainty of multicollinearity in the dataset. The weather data pixel size is ambiguous. The platform states that the coverage for each location ranges from 5 to 25 square kilometers. If the pixel size is 5 square kilometers, there will be no overlap in microclimate measurement coverage. If the pixel size is 25 square kilometers, the weather measurements will cover overlapping areas. The traffic volume data also introduces multicollinearity, since it is collected from 22 sensors along a 26.49-mile stretch of highway and carries 3 consecutive lags. Sensors next to one another measure similar traffic observations, so they may be highly redundant. The three consecutive lags may add further redundancy over time, since traffic volume in adjacent hours is autocorrelated, though not necessarily strongly enough to make the lags mechanically redundant.

Multicollinearity does not reduce the model’s predictive accuracy, since the overlapping signal is simply distributed across the correlated features without loss of information at the prediction stage. However, it compromises the interpretability of variable importance scores. When two or more features are highly correlated, any one of them can produce an equally good split, and the model selects among them effectively at random within each tree. The resulting importance score is divided arbitrarily across the correlated set rather than concentrated on a single variable, diluting the measured importance of each individual feature and obscuring which underlying signal is doing the predictive work.

To drop the multicollinear predictors, I use a block-wise approach similar to Kuhn and Johnson (2013), with two deliberate modifications: I restrict the selection process to positively correlated pairs and I use an outcome-aware tiebreak. First, I split the right-hand-side variables into two distinct blocks, each consisting of only traffic or weather variables. Within each block, I build a signed Pearson correlation matrix containing the correlations for every pairwise combination. I then walk through the upper triangle of the matrix and collect all strictly positive Pearson correlation coefficients greater than 0.9. Processing these pairs from strongest to weakest correlations, I drop the variable in each pair with the weaker association to the outcome, skipping any pair whose

members have already been removed. For example, if a 1-hour lagged traffic volume sensor is highly multicollinear with the same 2-hour lagged traffic sensor, the filter selects that pair. The filter then identifies the stronger predictor of accidents, keeps the variable, and trims the weaker predictor.

Strongly negative pairs are left in place even though a correlation near -1 carries as much redundancy as one near $+1$. Restricting the filter to positive pairs is a conservative choice. Inverse pairs such as lane occupancy and traffic speed can reflect distinct physical mechanisms, and dropping one member risks removing a distinct signal. The cost of this choice is some retained redundancy, which affects importance interpretation rather than predictive accuracy. After filtering under these rules, the process drops 130 variables, leaving a dataset of 102,959 observations across 196 variables.

4 Methodology

The methodological choices are oriented towards handling severe class imbalance in the distribution of accident hours in the data. Of the 102,959 hour-observations in the master dataset, only 2,696 (2.62%) contain a reported accident, while 100,263 (97.38%) do not. The practical consequence is that a degenerate model predicting “no accident” for every hour would achieve 97.38% accuracy without identifying a single real crash. Any model trained to minimize overall classification error will be heavily biased toward 0 and will systematically under-predict accident hours.

I independently implement a Probability Forest and an XGBoost machine learning algorithm to predict the hourly probability of an injury crash from traffic, weather, and temporal predictors. To obtain an out-of-sample measure of prediction accuracy, I split the data into a training set and a test set. The training set spans from January 1st, 2014, to January 1st, 2021, and the test set spans from January 1st, 2021, to September 30th, 2025. The test set contains 41,615 hourly observations with 1091 accident hours, making roughly a 60/40 split (59.6% and 40.4%) with an identical distribution of accident hours across the training and test splits (2.62% and 2.62%). I use the training set to tune each model’s parameters using an expanding-window cross-validation and evaluate each model’s accuracy on the held-out test set.

A standard classification Random Forest is a bagging tree-ensemble model that returns a predicted class label by majority vote across trees. My research design uses a Probability Forest, a variant that averages the trees’ leaf-class proportions into a continuous risk score between 0 and 1. This probability output lets me address class imbalance by selecting an operating threshold rather than relying on a fixed 0.5 cutoff. Variable importance is measured by the mean decrease in Gini impurity, identifying which predictors were most influential in determining accident risk.

An XGBoost model is a gradient-boosted tree ensemble. In contrast to the Probability Forest,

which grows its trees independently and averages them, XGBoost builds trees sequentially. Each new tree is fit to the gradient of the loss from the trees already in the ensemble, so that successive trees correct their predecessors’ errors. The model produces a continuous predicted probability of an injury crash, which is converted to a class label using a chosen operating threshold rather than a fixed 0.5 cutoff. Variable importance is measured by mean gain, which is the average improvement in the model’s objective that a split on a given feature contributes.

I tuned each model’s hyperparameters using expanding-window cross-validation, which preserves the chronological order of the data rather than randomly shuffling observations into folds as k-fold cross-validation does. The training window expands with each fold and never discards data from earlier years. The first fold trains on 2014–2016 and validates on 2017; the second trains on 2014–2017 and validates on 2018; this continues through the fourth fold, which trains on 2014–2019 and validates on 2020. The 2021 onward test set is never used during tuning.

Table 1: Selected hyperparameters and cross-validation performance by model

Parameter	Probability Forest	XGBoost
Trees (final model)	5,000	103
Max depth	4	4
Min node / leaf size	5 (<code>min.node.size</code>)	20 (<code>min_child_weight</code>)
Feature fraction	50% (<code>mtry = 141</code>)	80% (<code>colsample_bynode</code>)
Row fraction	70% (<code>sample.fraction</code>)	70% (<code>subsample</code>)
Regularization	Bootstrap + random <code>mtry</code>	L2 = 10.0 (<code>reg.lambda</code>)
Learning rate	—	0.05 (<code>eta</code>)
Class imbalance	Probability leaves	<code>scale_pos_weight = 37.22</code>
Importance metric	Gini impurity reduction	Mean gain
CV mean Youden’s J	0.2596 (± 0.0325)	0.2791 (± 0.0221)

Note: Hyperparameters are selected by expanding-window cross-validation over four folds (2014–2016 through 2014–2019 training windows). Both models train on a 283-column design matrix created by dummy expansion of the 196 retained variables, so the feature fractions apply to the expanded matrix: `mtry = 141 =  $\lfloor 0.5 \times 283 \rfloor$` . CV mean Youden’s J is the average of the four validation folds’ J statistics, with the standard deviation across folds in parentheses.

Youden’s J statistic is a single measure of how well a classifier separates the two outcomes, giving equal weight to correctly identifying accident and non-accident hours so a model cannot score well by simply predicting that accidents rarely occur. It ranges from 0 for chance performance to 1 for perfect separation. For each hyperparameter combination, I average the four folds’ Youden’s J scores, and the combination with the highest mean is selected. After tuning, each model is refit on

the entire seven-year training set using its selected hyperparameters (reported in Table 1) to grow a predictive tree structure; its predictive accuracy is then evaluated once on the held-out test set.

I assess each model’s performance through several metrics. The first is the area under the ROC curve (AUC) (Equation 1), which estimates how well the model predicts accidents on the data held out from training. The ROC curve plots the true positive rate against the false positive rate as the classification threshold sweeps from 1 to 0, and the AUC summarizes it as the probability that the model assigns a higher risk score to a randomly chosen accident hour than to a randomly chosen non-accident hour,

$$\text{AUC} = \int_0^1 \text{TPR}(f) df = \Pr(\hat{p}_i > \hat{p}_j \mid y_i = 1, y_j = 0) \quad (1)$$

where f indexes the false positive rate, $\text{TPR}(f)$ is the true positive rate at that false positive rate, $\hat{p}_i$ is the model’s predicted risk score for hour i , and y_i is its accident indicator. An AUC of 0.5 is equivalent to randomly guessing each outcome. Because AUC is threshold-independent, I select an operating threshold using the same Youden’s J criterion used in tuning (Equation 2), which corresponds to the point on the ROC curve lying farthest above the chance diagonal,

$$J = \max_{\tau} [\text{TPR}(\tau) - \text{FPR}(\tau)] \quad (2)$$

where τ is the candidate classification threshold. Selecting the operating threshold on the test-set ROC curve uses the held-out labels twice. The labels first trace the curve and then choose the point on it. The AUC is unaffected, but the at-threshold metrics depend on a threshold chosen with test-set information. As a robustness check, I re-select each model’s threshold by maximizing Youden’s J on pooled out-of-fold predictions from the four cross-validation folds and apply it frozen to the test set. The results section reports both operating points.

At this threshold, I report several outcomes drawn from the confusion matrix. Sensitivity (Equation 3) is the proportion of true accident hours correctly identified by the model,

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

where TP , FN , FP , and TN are the true positive, false negative, false positive, and true negative counts from the confusion matrix. The false positive rate (Equation 4) captures the percentage of non-accident hours the model incorrectly flags as high risk,

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4)$$

Precision (Equation 5) is the share of flagged hours that contain an actual accident, directly

relevant to the cost of a false alarm,

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

Lift over random (Equation 6) measures how many times more accidents the model captures per flagged hour than random patrol would, computed as sensitivity divided by the flagged rate,

$$\text{Lift} = \frac{\text{Sensitivity}}{(TP + FP)/N} \quad (6)$$

where N is the total number of test-set hours.

5 Results

Table 2 reports both models’ performance on the held-out test set, spanning January 1st, 2021, to September 30th, 2025, and containing 1,091 accident hours across 41,615 hourly observations.

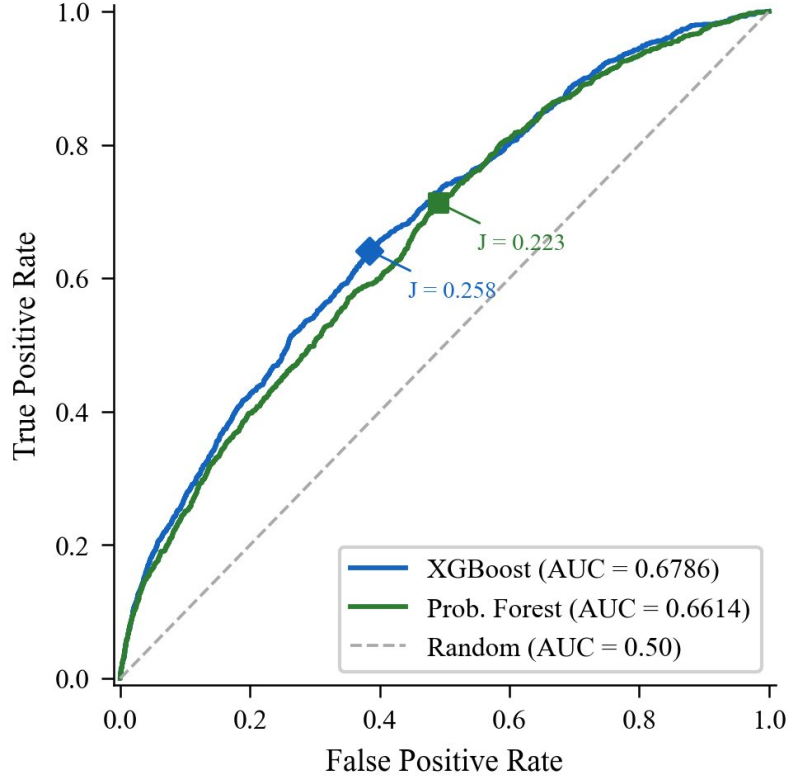
Table 2: Out-of-sample performance on the held-out test set, 2021–2025

Metric	Probability Forest	XGBoost
AUC	0.6614	0.6786
Sensitivity (TPR)	0.7131	0.6416
False positive rate	0.4904	0.3834
Specificity	0.5096	0.6166
Precision (PPV)	0.0377	0.0431
Youden’s J	0.2227	0.2582
Lift over random	1.437×	1.644×
Flagged hours	20,652	16,238
True positives	778	700
False positives	19,874	15,538
True negatives	20,650	24,986
False negatives	313	391
Test observations	41,615	41,615
Prevalence	0.0262	0.0262

Note: Bold marks the better value within each row. Threshold-dependent metrics are evaluated at each model’s Youden-optimal operating threshold (0.0260 for the Probability Forest and 0.4519 for XGBoost).

The Probability Forest achieves an out-of-sample AUC of 0.6614, and XGBoost achieves 0.6786. Both models rank a randomly chosen accident hour above a randomly chosen non-accident hour roughly two-thirds of the time, a noticeable improvement above the 0.5 random-guessing benchmark. XGBoost’s higher AUC indicates it distinguishes between accident and non-accident hours more reliably across the full range of thresholds. Figure 2 plots both ROC curves against the random-guessing diagonal, marking each model’s Youden-optimal operating point.

Figure 2: ROC curves on the held-out test set with Youden-optimal operating points



Applying Youden’s J to each model’s ROC curve yields the operating threshold for that model. The Probability Forest’s maximized J statistic is 0.2227, and XGBoost’s is 0.2582. The thresholds place the two models at noticeably different operating points: the forest flags 20,652 of the 41,615 test hours as high risk (49.6%), while XGBoost flags 16,238 (39.0%).

The two operating points produce a clear trade-off between detection and false alarms. The Probability Forest achieves a sensitivity of 0.7131, correctly identifying 778 of the 1,091 accident hours. The XGBoost achieves a sensitivity of 0.6416, capturing 700. The forest’s higher capture rate comes at the cost of a higher false positive rate: it incorrectly flags 49.04% of non-accident hours, compared to 38.34% for XGBoost.

Precision is low for both models, which is expected given the 2.62% prevalence of accident hours. The Probability Forest’s precision is 0.0377, meaning that 1 in every 27 flagged hours contains

an actual accident, while XGBoost’s precision is 0.0431, or 1 in every 23 flagged hours. Because accidents are so rare, overall accuracy is a poor measure of performance here, since a model that flagged nothing would still appear highly accurate even if it identified no crashes. Lift over random contextualizes the models’ accuracy in terms of the difficulty of the prediction task. The Probability Forest captures accidents at 1.44 times the rate of random patrol per flagged hour, while XGBoost captures them at 1.64 times that rate. Both models concentrate accident risk into their flagged hours, with XGBoost doing so more efficiently.

Comparing each model’s results does not produce a clearly superior specification. The XGBoost wins on ranking ability, targeting efficiency, and false-alarm control. The model flags roughly 4,400 fewer hours while achieving higher precision and lift. The Probability Forest outperforms XGBoost on raw detection, capturing 78 more accident hours. Which set of predictions is preferable depends on how an agency weighs a missed accident against an unnecessary deployment, which is precisely the question the cost-benefit analysis is constructed to answer.

Figures 3 and 4 display each model’s confusion matrix on the test set. The Probability Forest correctly flags 778 accident hours (true positives) and misses 313 (false negatives), while flagging 19,874 non-accident hours as high risk (false positives) and correctly leaving 20,650 unflagged (true negatives). XGBoost correctly flags 700 accident hours and misses 391, with 15,538 false positives and 24,986 true negatives. The four cells highlight the tradeoff between the two models: the Probability Forest converts 78 of XGBoost’s misses into detections, but pays for them with 4,336 additional false alarms, roughly 56 extra flagged hours for each additional accident captured.

The operating thresholds above are selected on the test set itself, so I check whether the results survive a threshold frozen before the test period. I re-select each model’s threshold by maximizing Youden’s J on pooled out-of-fold predictions from the four cross-validation folds and apply it unchanged to the test set. Table 3 compares the two operating points. The forest’s sensitivity falls from 0.7131 to 0.6627 and it flags 1,664 fewer hours. XGBoost moves the opposite way. Its sensitivity rises from 0.6416 to 0.6746 and it flags 1,804 more hours. Under frozen thresholds the two models converge to similar operating points and XGBoost captures more crash-hours than the forest. Part of the headline detection gap therefore reflects test-set threshold selection rather than a difference in model quality. The economic conclusion is unchanged. At the central 9 percent crash reduction, the one-officer benefit-cost ratios under frozen thresholds are 10.1 for the forest and 10.8 for XGBoost.

Figure 3: Probability Forest confusion matrix on the test set (Youden-optimal threshold = 0.0260)

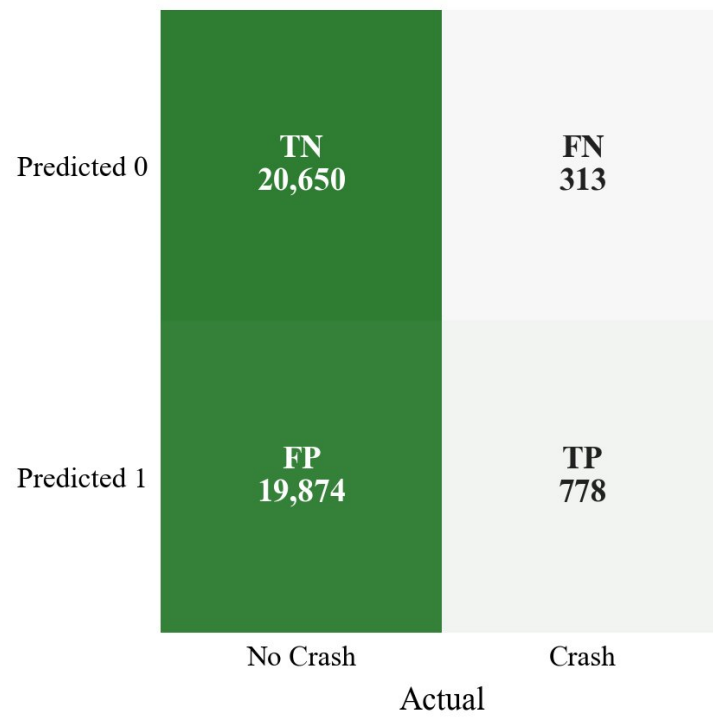


Figure 4: XGBoost confusion matrix on the test set (Youden-optimal threshold = 0.4519)

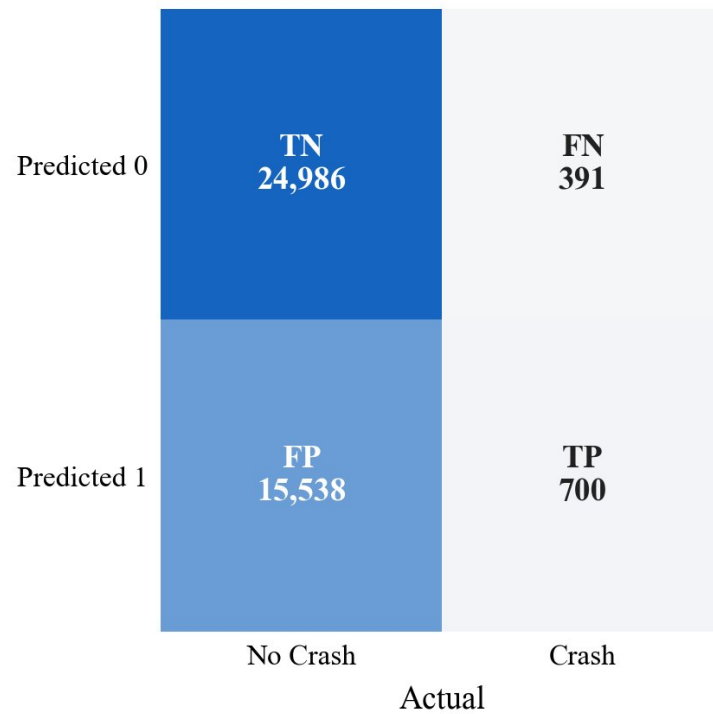


Table 3: Operating points under test-set-selected and cross-validation-selected thresholds

Metric	Probability Forest		XGBoost	
	Test-set	CV-frozen	Test-set	CV-frozen
Threshold	0.0260	0.0274	0.4519	0.4363
Sensitivity	0.7131	0.6627	0.6416	0.6746
False positive rate	0.4904	0.4507	0.3834	0.4271
Flagged hours	20,652	18,988	16,238	18,042
True positives	778	723	700	736
False positives	19,874	18,265	15,538	17,306
True negatives	20,650	22,259	24,986	23,218
False negatives	313	368	391	355

Note: CV-frozen thresholds maximize Youden’s J on pooled out-of-fold predictions from the four expanding-window cross-validation folds and are applied unchanged to the test set. Test-set columns reproduce the operating points in Table 2.

Figures 5 and 6 display the top 20 variables by importance for each model. Despite measuring importance on different bases, the two models’ most important features considerably overlap. The top 20 predictors in both models are split evenly between weather and traffic variables. The weather signal is concentrated overwhelmingly in rain. Summit precipitation variables rank at or near the top of both lists. XGBoost’s two highest-gain features are the summit’s precipitation type and rain indicator. The Probability Forest places summit precipitation, rain indicators, and precipitation type in positions two through four. Santa Cruz-side precipitation measures appear in both models, consistent with rain arriving off the coast and over the summit before reaching the drier Santa Clara side, where the corresponding precipitation variables rank 16th in the XGBoost and drop out of the top 20 entirely in the Probability Forest.

The traffic signal is equally consistent. Every traffic variable in both top 20 lists is a 1-hour lag; no 2- or 3-hour lag appears in either ranking. The results indicate that the most recent available flow and occupancy conditions carry nearly all of the traffic-based predictive signal. The Probability Forest’s single most important variable is the 1-hour lagged total flow at sensor 400416, with sensors 405281, 403329, 400240, 400298, and 400715 appearing in both models’ rankings through some combination of their lagged flow, occupancy, or speed measures. This suggests that a specific subset of locations along the corridor is informative across the model families. The Probability Forest also assigns importance to Santa Clara sea-level pressure, in both its current and 24-hour-lagged forms, mirroring the atmospheric pressure findings of Wang et al. (2024) and Yan and Shen (2022).

Figure 5: Probability Forest: top 20 variables by mean decrease in Gini impurity

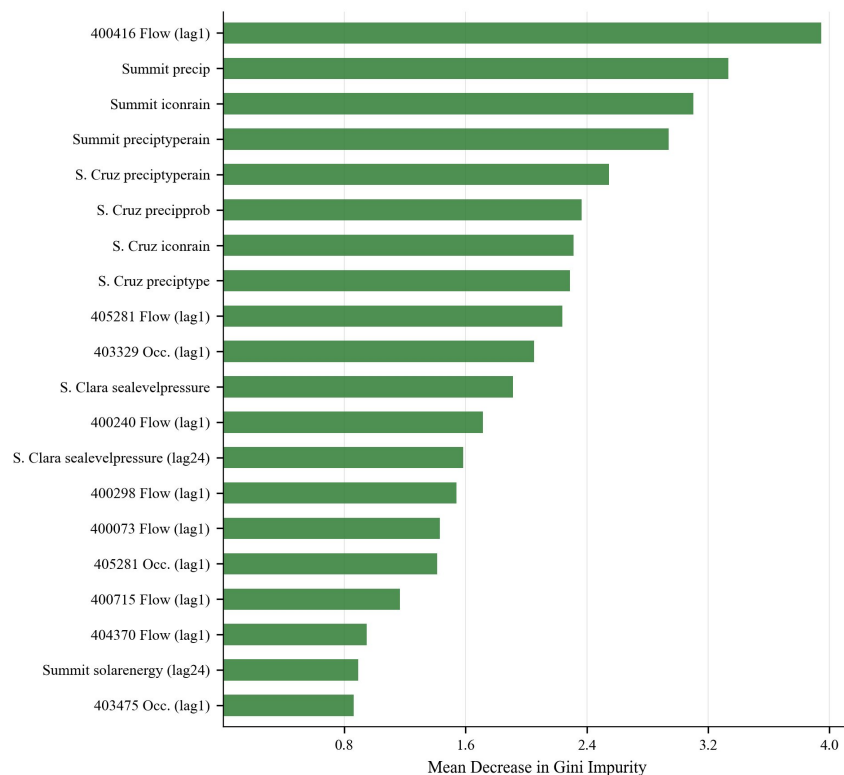
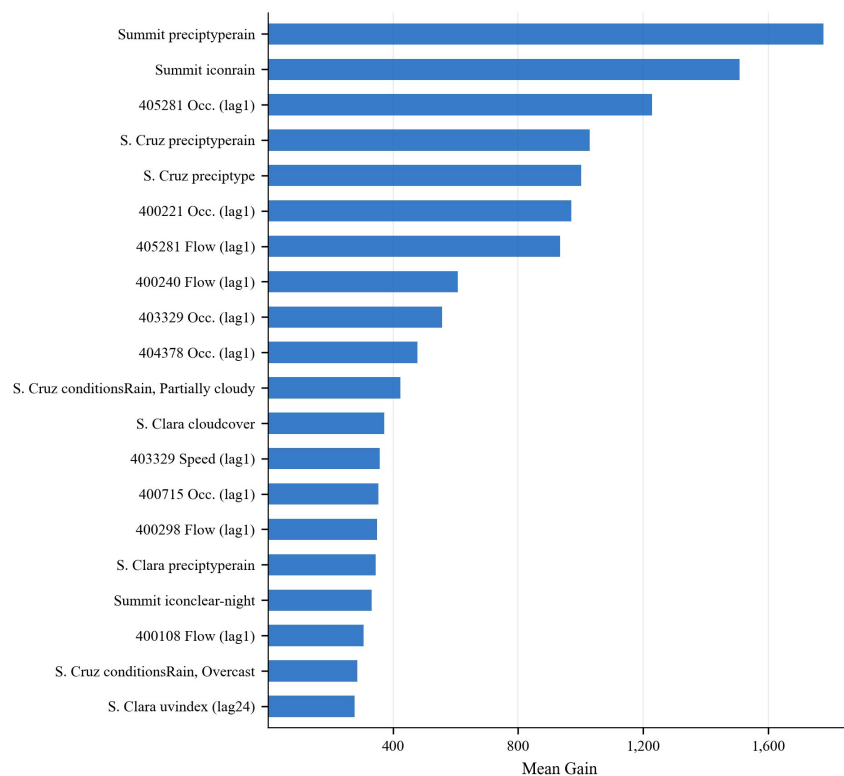


Figure 6: XGBoost: top 20 variables by mean gain



6 Cost-Benefit Analysis

The predictive models gain value only when they are paired with a real-world intervention that translates predictions into fewer crashes. This section asks whether targeted CHP enforcement during model-identified high-risk hours is economically justified. I use that framing to select the preferred model specification. The analysis proceeds in three parts. The first part builds the benefit side. It develops the benefit formula and states the assumptions and deterrence mechanism behind it. It then estimates the dollar value of the crashes each model can deter. The second part builds the cost side from a 2024 CHP cost proposal scaled to the corridor. The third part combines the two into benefit-cost ratios and net benefits and presents the Probability Forest and XGBoost results for each criterion.

6.1 Benefit Framework

The annual program benefit is the product of three components.

$$\text{Annual Benefit} = C_{\text{baseline}} \times \rho_{\text{capture}}^m \times (1 - \text{CMF}) \quad (7)$$

Here, C_{baseline} is the annualized social cost of injury crashes on the corridor. The capture share ρ_{capture}^m is the fraction of that cost occurring during model-flagged hours, where m is either the Probability Forest or XGBoost. CMF is the crash modification factor. It gives the multiplier on crash frequency under the intervention, so that $(1 - \text{CMF})$ is the proportional reduction in crash frequency. All monetary values are expressed in 2024 dollars to match the USDOT monetization schedule’s dollar-year.

6.2 Key Assumptions

The framework rests on five main assumptions. First, the accident severity is uniformly distributed across crash-hours. A flagged crash-hour carries the same expected severity as an unflagged one. This assumption is conservative because rain and congestion drive the models’ risk scores and are likely positively correlated with crash severity. If the models’ flagged accidents are, on average, more severe, the models’ true benefits are higher than this paper estimates. Second, the Phillips, Ulleberg, and Vaa (2011) crash-reduction estimate, introduced in the next subsection, transfers to a California mountain highway. The meta-analysis spans heterogeneous road types and enforcement contexts, so transferring the findings to a specific highway carries uncertainty. Third, the CHP can deploy during all flagged hours and cover the corridor while deployed. The benefit formula credits the full deterrent effect to every flagged crash-hour. When coverage falls short, deterrence reaches only part of the flagged risk. Realized benefits then fall below the estimates reported

here. This assumption therefore adds upward bias to the estimated benefits. Adding officers raises coverage toward the assumed full-deterrence case; the deployment caveat below returns to this point. Fourth, the analysis assumes that both benefits and costs scale linearly with the number of officers added to the deployment response. Finally, the operating thresholds behind each model’s capture rate come from the test-set ROC curves. Re-selecting each threshold on pooled cross-validation predictions and freezing it shifts the forest’s capture rate down and XGBoost’s up (Table 3), while the one-officer central-estimate benefit-cost ratios remain above 10 for both models.

6.3 Enforcement Mechanism and the Crash-Reduction Effect

The benefits are solely a function of deterrence. Drivers who perceive a higher risk of enforcement contact modify their behavior during flagged hours. They reduce speed, increase following distance, and avoid impaired driving. The deterrent effect applies only while officers are deployed and dissipates outside those hours. Only the flagged share of social cost is therefore subject to reduction.

The magnitude of the effect comes from Phillips, Ulleberg, and Vaa (2011). This meta-analysis, published in *Accident Analysis and Prevention*, synthesizes 119 accident-effect estimates from 67 evaluation studies conducted across 12 countries between 1975 and 2007. It is the appropriate source here because the standard federal catalog of crash-reduction estimates covers only physical engineering countermeasures and offers no factor for high-visibility enforcement. The meta-analysis is the only study that directly addresses my proposed intervention. Its evidence base consists of road safety campaigns, which the paper defines to include accompanying police enforcement. Its meta-regression isolates the delivery features that yield the largest reductions in accidents.

The weighted average across all 119 estimates is a 9 percent reduction in accidents, with a 95 percent confidence interval of 6 to 12 percent (a CMF of 0.91, with bounds of 0.94 and 0.88). This figure averages across campaigns of all types, most of which rely on mass media rather than enforcement. The authors find that accompanying enforcement, roadside delivery, personal communication, and proximity to the targeted behavior are each associated with larger reductions. In the subgroup comparison, campaigns paired with enforcement average a 13 percent reduction, compared with 10 percent for those without.

A targeted CHP patrol concentrated on high-risk hours is an enforcement-based, roadside intervention delivered close to the behavior it deters, placing it in the category of interventions tied to above-average effectiveness. The 9 percent central estimate is therefore conservative for this application. I evaluate Equation 7 at 6, 9, and 12 percent reductions for the lower bound, central estimate, and upper bound.

6.4 Baseline Social Cost

The baseline social cost applies USDOT comprehensive crash cost values to the held-out Highway 17 collision record. Comprehensive crash costs capture the full social burden of a crash. They cover emergency medical care, lost productivity, and property damage. They also cover emergency response and the monetized value of quality-of-life loss from willingness-to-pay studies. The KABCO scale (Killed, A-injury, B-injury, C-injury, O-property damage, U-unknown) maps directly to TIMS/SWITRS collision severity codes 1-4. USDOT reports unit values of \$13,700,000 per fatality and \$1,302,300 per suspected serious injury. The minor-injury and complaint-of-pain values are \$256,300 and \$122,400 (U.S. Department of Transportation 2026; Blincoe et al. 2023). Table 4 reports the full KABCO mapping and unit values. The record contains 1,143 individual collisions across 1,091 distinct crash-hours, because some hours contain multiple crashes. Applied to these 1,143 collisions over the 4.75-year test window, the values yield an annualized social cost of \$113,896,653. Fatal collisions account for only 1.5 percent of the total count but contribute roughly 43 percent of the total social cost. About 3.6 fatal crashes per year drive this share, so the program’s benefits hinge on preventing them. Table 5 reports per-severity counts, annual rates, and annualized costs.

Table 4: KABCO injury-severity scale, mapping to TIMS/SWITRS collision-severity codes, and per-injury monetization values from USDOT BCA Guidance

KABCO	Description	TIMS Severity Code	Unit Value (2024\$)
K	Fatal injury	1 – Fatal injury	13,700,000
A	Incapacitating injury	2 – Suspected serious (severe) injury	1,302,300
B	Non-incapacitating injury	3 – Suspected minor (visible) injury	256,300
C	Possible injury	4 – Possible injury (complaint of pain)	122,400
O	Property damage only	— (excluded from TIMS extract)	5,500
U	Injury, severity unknown	— (not used in this analysis)	238,500

Sources: *Benefit-Cost Analysis Guidance for Discretionary Grant Programs* (U.S. Department of Transportation, Office of the Secretary 2025); underlying valuations from Blincoe et al. (2023). TIMS severity code definitions per UC Berkeley SafeTREC Transportation Injury Mapping System (SWITRS dictionary).

Table 5: Highway 17 collisions by severity, January 1, 2021 through September 30, 2025 ($N = 1,143$ collisions over 4.75 years)

Severity	Count	Pct (%)	Annual Rate	Unit Value (\$)	Annual Cost (\$)
Fatal (K)	17	1.5	3.58	13,700,000	49,046,000
Suspected Serious (A)	91	8.0	19.18	1,302,300	24,978,114
Suspected Minor (B)	467	40.9	98.41	256,300	25,222,483
Complaint of Pain (C)	568	49.7	119.69	122,400	14,650,056
Total	1,143	100.0	240.86	—	113,896,653

Source: Author’s calculations from TIMS/SWITRS Highway 17 collision data, 2021–2025. Property-damage-only collisions are excluded per TIMS data convention. Annual rates use the exact window length (January 1, 2021 through September 30, 2025, about 4.745 years) and are rounded to two decimals. Each annual cost equals the displayed annual rate multiplied by the KABCO unit value from Table 4, expressed in 2024 dollars.

6.5 Capture Rate

CHP deploys only during model-flagged hours, so not every corridor crash is deterrable. The relevant adjustment is each model’s sensitivity to crash hours. The Probability Forest identifies 778 of the 1,091 crash-hours in the test set.

$$\rho_{\text{capture}}^{\text{PF}} = \frac{778}{778 + 313} = 0.7131 \quad (8)$$

XGBoost identifies 700 of the 1,091 crash-hours.

$$\rho_{\text{capture}}^{\text{XGB}} = \frac{700}{700 + 391} = 0.6416 \quad (9)$$

The capture rate measures crash-hours, while social cost sums over the 1,143 individual collisions. The uniformity assumption reconciles the two. Under that assumption, Probability Forest-flagged hours contain 71.31 percent of annualized social cost, about \$81.2 million. XGBoost-flagged hours account for 64.16 percent, or about \$73.1 million. The remaining shares fall outside deployed hours, so the analysis excludes them from the deterrable benefit. Table 6 summarizes each model’s capture of crash-hours and the flagged share of social cost.

Table 6: Crash-hour capture and flagged social cost by model, held-out test set

Model	Captured	Missed	Capture Rate	Flagged Cost (\$/yr)
Probability Forest	778	313	0.7131	81,220,528
XGBoost	700	391	0.6416	73,077,596

Note: Captured and missed are counts of the 1,091 test-set crash-hours. The capture rate is each model’s sensitivity over crash-hours (Equations 8 and 9). Flagged cost applies the unrounded capture rate (778/1,091 and 700/1,091) to the \$113,896,653 annualized corridor cost from Table 5 under the uniform-severity assumption.

6.6 Annual Benefit

Combining the three components yields the annual program benefit. For the Probability Forest, the benefit is about \$4.87 million at the lower bound, \$7.31 million at the central estimate, and \$9.75 million at the upper bound. XGBoost yields \$4.38 million, \$6.58 million, and \$8.77 million. The forest’s higher capture rate produces roughly 11 percent more deterrable benefit at every effectiveness level. Table 7 reports the full benefit schedule.

Table 7: Annual program benefits by model and crash-reduction effect (2024 dollars)

CMF Bound	Crash Reduction	Probability Forest (\$)	XGBoost (\$)
Lower bound (CMF = 0.94)	6%	4,873,232	4,384,656
Central (CMF = 0.91)	9%	7,309,848	6,576,984
Upper bound (CMF = 0.88)	12%	9,746,463	8,769,312

Note: Annual benefit = flagged annualized social cost (Table 6) $\times$ (1 – CMF), per Equation 7. CMF bounds are from Phillips, Ulleberg, and Vaa (2011). Benefits are constant across deployment intensities because the underlying meta-analysis does not vary effect sizes by patrol density.

6.7 Program Costs

The cost side uses a 2024 CHP cost proposal to the Santa Barbara County Association of Governments for supplemental enforcement on State Route 166 (Santa Barbara County Association of Governments 2024). This proposal is the most directly comparable California pricing precedent available. No equivalent published pricing exists for Highway 17. The proposal bills officer overtime at \$122.51 per hour, sergeant overtime at \$149.05 per hour, and patrol vehicle costs at \$1.45 per mile (Santa Barbara County Association of Governments 2024). It supplies two further parameters for scaling the program to Highway 17: a supervisor-to-officer ratio of about 1 to 8.6, or 140 sergeant hours per 1,200 officer hours, and a vehicle utilization rate of 20 miles per officer-hour.

The billed rates are CHP overtime pay rates plus a per-mile vehicle charge. They measure what a sponsoring agency would pay under such an agreement rather than CHP’s full departmental cost of providing the service. Unbilled overhead such as dispatch and administration would raise true costs above this basis and lower the reported ratios modestly.

Each model’s flagged hours are the clock hours during which CHP would deploy. These hours equal the sum of true positives and false positives. The Probability Forest flags 20,652 hours over the 4.75-year window. This corresponds to 20,652 officer-hours, 2,409 sergeant-hours, and 413,040 vehicle-miles. The one-officer program cost is about \$3.49 million, or \$734,000 annualized. XGBoost flags 16,238 hours. This corresponds to 16,238 officer-hours, 1,894 sergeant-hours, and 324,760 vehicle-miles. The one-officer cost is about \$2.74 million, or \$577,000 annualized. The forest’s flagged hours represent about 2,582 eight-hour shifts, roughly 10.4 per week across the window. XGBoost represents about 2,030 shifts, roughly 8.2 per week. Costs scale linearly with officer count. Table 8 reports the cost breakdown by model.

Table 8: One-officer program costs by model over the 4.75-year test window (2024 dollars)

Line Item	Probability Forest	XGBoost
Flagged clock-hours (officer-hours)	20,652	16,238
Sergeant-hours	2,409	1,894
Vehicle miles	413,040	324,760
Officer cost (\$122.51/hr)	2,530,077	1,989,317
Sergeant cost (\$149.05/hr)	359,061	282,301
Vehicle cost (\$1.45/mi)	598,908	470,902
Total program cost	3,488,046	2,742,520
Annualized cost	734,325	577,373

Source: Santa Barbara County Association of Governments (2024), Attachment B. Sergeant hours follow the implied 1:8.6 supervisor-to-officer ratio (140 sergeant hours per 1,200 officer hours); vehicle miles follow the implied 20 miles per officer-hour, with one patrol vehicle per officer. The annualized cost divides total cost by the 4.75-year window. Table 9 reports the annualized cost at each intensity.

6.8 Results and Model Comparison

The benefit-cost ratio (BCR) is the ratio of annual benefits to annualized costs. Benefits remain constant across deployment scenarios because the underlying meta-analysis does not vary effect sizes by patrol density. Costs scale linearly with officer count. Each model therefore produces a 12-cell BCR matrix across four deployment intensities and three crash-reduction bounds. Table 9

reports both matrices. Probability Forest BCRs run from 1.66 at four officers and the 6 percent lower bound to 13.27 at one officer and the 12 percent upper bound. Central-estimate values run from 2.49 to 9.95. XGBoost BCRs run from 1.90 to 15.19, with central-estimate values from 2.85 to 11.39. All 24 cells exceed the conventional threshold of 1.0. Targeted enforcement is therefore economically justified under either model across the full range of plausible deployment intensities and crash-reduction effects.

Table 9: Benefit-cost ratios by model, deployment intensity, and crash-reduction effect

Deployment	Annualized Cost (\$)	BCR (6%)	BCR (9%)	BCR (12%)
<i>Probability Forest</i>				
1 officer per shift	734,325	6.64	9.95	13.27
2 officers per shift	1,468,651	3.32	4.98	6.64
3 officers per shift	2,202,976	2.21	3.32	4.42
4 officers per shift	2,937,302	1.66	2.49	3.32
<i>XGBoost</i>				
1 officer per shift	577,373	7.59	11.39	15.19
2 officers per shift	1,154,745	3.80	5.70	7.59
3 officers per shift	1,732,118	2.53	3.80	5.06
4 officers per shift	2,309,491	1.90	2.85	3.80

Note: BCR = annual benefit (Table 7) / annualized cost (Table 8). All 24 model-scenario-bound combinations exceed the conventional BCR threshold of 1.0.

The comparison between the two models depends on the decision criterion. On the ratio criterion, XGBoost dominates. Its BCR exceeds the Probability Forest’s in every cell by a constant factor of about 1.14. That factor equals the ratio of the two models’ lifts (1.644/1.437) because the BCR tracks the captured benefit per flagged hour. A model that concentrates crash risk into fewer and more precise flags produces a higher BCR. XGBoost does exactly that. On the net-benefit criterion, the ranking mostly reverses. At the central 9 percent crash reduction, the Probability Forest’s annual net benefit exceeds XGBoost’s at every deployment intensity. The gap runs from \$6.58 million against \$6.00 million at one officer to \$4.37 million against \$4.27 million at four officers. The forest captures 78 additional crash-hours. These generate about \$733,000 of additional annual benefit at the central estimate. The additional annualized cost runs from \$157,000 to \$628,000 depending on deployment intensity. The marginal flagged hours therefore pay for themselves in every cell except one. That cell is four officers at the 6 percent lower bound, where the incremental benefit of \$489,000 falls short of the incremental cost. USDOT guidance treats the BCR as the metric for ranking competing projects under a fixed budget. It treats net benefit as the measure of

total social value created. Under a binding enforcement budget, XGBoost targeting is preferable. An agency that wants to maximize total averted social cost deploys on the forest’s wider flag set at low to moderate officer counts.

6.9 Deployment Caveat

The BCR declines mechanically with officer count because the analysis holds benefits constant while costs scale linearly. That treatment follows from the Phillips, Ulleberg, and Vaa (2011) meta-analysis, which pooled effect sizes across heterogeneous campaigns without separating them by patrol density. Two readings are defensible. At low deployment intensity, the program may face a capacity constraint. A single officer cannot patrol the full 26-mile corridor continuously during all flagged hours, so adding officers raises coverage and absolute benefits. At high deployment intensity, deterrence likely shows diminishing returns. The marginal deterrent effect of a fourth officer is likely smaller than that of the first. The conservative reading treats the one- and two-officer scenarios as closest to the deployment intensity studied in the underlying literature. These scenarios represent the most defensible operating points.

6.10 The Undirected-Patrol Benchmark and Model Contribution

An undirected-patrol benchmark measures the economic value the models add over untargeted deployment. The benchmark places a single CHP unit on the corridor during every hour, with no model guiding deployment. The 4.75-year test window contains 41,615 hours. The combined billed rate is \$168.90 per officer-hour. This rate combines \$122.51 in officer overtime, \$17.39 in supervision at the 1-to-8.6 ratio, and \$29.00 in vehicle cost at 20 miles per hour. Continuous single-unit coverage therefore costs about \$7.03 million over the window, or \$1.48 million annualized. A unit present every hour captures every crash-hour. The benefit then equals the full corridor social cost multiplied by the crash reduction. That benefit ranges from \$10.25 million per year at the central 9 percent effect to \$6.83 million at the 6 percent lower bound and \$13.67 million at the 12 percent upper bound. The benefit-cost ratio is 6.9 at the central estimate, within a range of 4.6 to 9.2. Annual net benefit reaches \$8.77 million at the central estimate. Table 10 summarizes the benchmark. Blanket patrol of Highway 17 is therefore strongly justified before any targeting. The social cost of corridor crashes so far exceeds the enforcement cost that even the least efficient deployment returns several dollars for every dollar spent.

Table 10: Undirected-patrol benchmark: continuous single-unit coverage of all 41,615 test-window hours (2024 dollars)

CMF Bound	Crash Reduction	Annual Benefit (\$)	BCR
Lower bound (CMF = 0.94)	6%	6,833,799	4.6
Central (CMF = 0.91)	9%	10,250,699	6.9
Upper bound (CMF = 0.88)	12%	13,667,598	9.2

Note: The combined billed rate is \$168.90 per officer-hour (\$122.51 officer overtime, \$17.39 supervision at the 1:8.6 ratio, and \$29.00 vehicle cost at 20 miles per hour). Continuous coverage of 41,615 hours costs about \$7.03 million over the window, or \$1,479,742 annualized. A unit present every hour captures every crash-hour, so the benefit applies the crash reduction to the full \$113,896,653 annualized corridor cost. BCR = annual benefit / annualized cost.

Both benefit and cost scale linearly with the hours patrolled. The undirected ratio is therefore invariant to coverage. Patrolling every hour and patrolling any random subset both return a benefit-cost ratio of about 6.9 at the central estimate. Enforcement is viable before any model enters the picture. The models add value by concentrating the same enforcement dollars on hours with disproportionate crash risk. Random deployment over the forest’s 49.6 percent of test hours captures 49.6 percent of crash-hours, while the forest captures 71.31 percent. That gain is a 1.44x lift. Random deployment over XGBoost’s 39.0 percent of hours captures 39.0 percent, while the model captures 64.16 percent. That gain is a 1.64x lift. Each model multiplies the undirected return by its lift factor. This raises the central benefit-cost ratio from 6.9 to 9.95 and 11.39 in the one-officer central-estimate cells reported above. The models therefore improve the return on an already-justified program by 44-64 percent.

7 Conclusion

This paper asked whether machine learning can predict injury crashes on California Highway 17 and whether those predictions justify targeted enforcement. To answer it, I assembled an 11.75-year hourly dataset combining CHP injury-crash records, traffic measurements from 22 corridor sensors, and weather observations from three distinct microclimates along the route. I trained a Probability Forest and an XGBoost model on seven years of data and evaluated both on a 4.75-year held-out window.

Both models predict well enough to act on. The Probability Forest achieves an AUC of 0.6614 and captures 778 of the 1,091 crash hours in the test set. XGBoost achieves an AUC of 0.6786 and captures 700 while flagging roughly 4,400 fewer hours. Each model ranks a randomly chosen crash hour above a non-crash hour about two-thirds of the time and concentrates crash risk into

its flagged hours, at 1.44 and 1.64 times the random patrol rate. The two models agree on what drives that risk. Summit rain dominates the weather signal. One-hour-lagged flow and occupancy at a small set of sensor locations carry nearly all of the traffic signal.

The cost-benefit analysis converts those predictions into a deployment decision. Crashes on this 26.49-mile corridor destroy roughly \$114 million in social value per year. A loss that severe means nearly any credible enforcement response clears a benefit-cost ratio of one. Model-directed deployment improves on that baseline by 44-64 percent, lifting the central one-officer ratio to 9.95 under the Probability Forest and 11.39 under XGBoost. All 24 cells in the deployment matrix exceed 1, and neither model dominates the other. XGBoost extracts more value per flagged hour and suits an agency with a smaller budget. The Probability Forest captures more crashes and yields a higher total net benefit across most deployment intensities.

Two limitations matter most. The coverage assumption credits the full deterrent effect to every flagged hour, yet a single unit cannot blanket 26 miles of mountain highway for an entire flagged hour. Also, since the model relies on the most recent hour of traffic data and on current weather, the CHP deployment needs to be readily available. Realized benefits will fall short of estimates until deployment intensity matches the flagged hours. The second limitation is operational. The models predict from lagged sensor and weather data that exists cleanly in a historical file. A working program requires a live pipeline that uses current traffic data and forecast weather for the coming hour, tolerates sensor outages, and issues each forecast before the hour begins.

The evidence supports action now. Every model, deployment intensity, and crash-reduction estimate in this paper returns more in averted social cost than the program spends. Highway 17 averages roughly 3.6 fatal crashes per year, and each year without a heightened response forfeits millions of dollars in preventable harm. CHP and the regional transportation agencies should fund a one- or two-officer pilot guided by either model, build the real-time data pipeline the program requires, and measure crash outcomes against the corridor's baseline. A pilot would replace the transferred meta-analysis estimate with corridor-specific evidence and settle the one question this paper cannot: how much deterrence these deployments deliver in practice.

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